

RESEARCH ARTICLE

Fuzzy Logic Recommender Model for Housing

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ABSTRACT Recommending suitable housing implies significant challenges owing to the continuous increase in demand and the need to meet habitability standards. This document presents an innovative approach with which to address these challenges through the use of a housing recommendation method based on distances to key spatial points and the latent characteristics of the properties. The proposed method employs objective distances from the properties to points of interest, such as educational centers, medical centers, pharmacies, shops, entertainment, 911 security cameras and public transport stations. These distances are calculated on the basis of the area in which the property is located, thus providing an accurate assessment of the environment. Moreover, housing features are grouped into three correlated latent factors: Size and Value, Environment and Comfort, and Age and Safety. The recommendation system relies on fuzzy control to manage user preferences and select appropriate input data with which to test the model. A content-based filtering approach is used initially, as housing ratings are unavailable. The model predicts a percentage of membership in each cluster, which makes it possible to handle uncertainty by offering properties from different groups in a proportional manner. Euclidean distance is employed in order to measure the similarity between user preferences and housing characteristics, after which the search time is optimized by utilizing metaheuristic methods, of which the bat algorithm provides the best performance in terms of time. This algorithm selects the properties displayed to the user on the basis of natural features extracted from real estate platforms by means of web scraping techniques. The system is built with a Model-View-Controller architecture using Python, Flask, and SQLite. Personal customer data is also recorded in order to create clusters and calculate distances for new customers, thus allowing properties with high ratings to be recommended. This approach combines collaborative and content-based filtering, creating a hybrid system that improves recommendation accuracy and relevance. This analysis shows that the new recommendation method is an effective and accessible solution with which to select suitable housing.

INDEX TERMS Recommender systems, fuzzy logic, adequate housing.

I. INTRODUCTION

Access to adequate housing is a global challenge affecting millions of people worldwide. According to United Nations (UN) data, more than 1.6 billion people live in inadequate housing conditions, while another 900 million simultaneously reside in informal settlements or camps, both in developed and developing nations. These figures underscore the urgent need to address the housing problem both comprehensively and effectively [1]. In this context,

recommendation systems (RS) have appeared as an innovative paradigm designed to detect citizens' desires and preferences in a personalized manner through the use of predictive models, which is seen as a solution that would support users' decision-making.

The literature in the housing sector field has proposed several RS techniques [2], which have focused on prices in the context of housing [3]. Some of the techniques used are collaborative filtering [4], content-filtering [5], matrix factorization [3] and hybrid recommender systems [6]. However, their application has focused on predicting and estimating prices and recommendations on the basis of specific housing

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characteristics, without considering the real needs of citizens beyond price, but rather the set of elements that constitute dignified housing.

The characteristics defining adequate housing that are outlined by the UN include aspects such as: 1. Security of tenure; 2. Availability of services, materials, facilities and infrastructure; 3. Affordability; 4. Habitability; 5. Accessibility; 6. Location, and 7. Cultural adequacy.

Uncertainty in various forms, such as vagueness and imprecision, is a common theme in the research presented. Fuzzy topological spaces represent uncertainty and vagueness within systems and are applied in computing, engineering and economics [7]. Fuzzy logic, which can handle uncertain and imprecise information, is used for decision-making processes and data mining, particularly in educational settings for the purpose of linguistic assessment [2]. Measuring uncertainty in fuzzy-valued information systems provides a new perspective for data analysis, as it explores the intrinsic properties of uncertainty in these systems by means of distance calculations and tolerance relations [3]. Moreover, fuzzy probabilistic information systems combine fuzzy relations that satisfy probability distributions, providing a unique approach with which to managing uncertainty in information systems [4]. The study of cooperative games within organizations takes into account uncertainty regarding benefits, especially in peer-group situations, leading to modeling games with fuzzy uncertainty and solution concepts based on game theory [5].

Uncertainty is inherent in many real-world problems, in the case of which decisions and predictions must be made on the basis of incomplete, imprecise, or vague data. In this context, fuzzy logic is a powerful tool with which to handle this uncertainty in an effective manner. Unlike classical logic, which requires binary decisions (true or false), fuzzy logic makes it possible to work with degrees of truth, thus providing a more flexible representation that is closer to the reality of human knowledge and perception [8]. This capability is instrumental in recommendation systems, in which user preferences and item characteristics are only sometimes clear or definitive [9]. Fuzzy logic has also been applied successfully in various fields, from industrial control systems to artificial intelligence, demonstrating its effectiveness as regards managing uncertainty and making decisions [10]. For example, in housing recommendations, attributes such as “comfort” or “favorable location” may be subjective and vary in degrees, something that fuzzy logic can handle more intuitively than traditional approaches [11], [12]. The objective of this research is, therefore, to present a Spatial Fuzzy C-Means Clustering (FCM) Algorithm model [13] focused on the challenges of housing planning in the context of the UN. Furthermore, it identifies citizens’ needs and preferences and provides personalized recommendations by considering various factors and key elements of adequate housing. The specific objective of our study is to contribute to achieving the following objectives, which are:

- 1) To collect raw data by means of web scraping and subsequently applying preprocessing techniques in order to remove outliers, thus enabling housing clustering methods to be used effectively.
- 2) In the processed dataset, to calculate the distances between houses and specific places organized by parish; to reduce the dimensions of variables, and to create groups of records using an unsupervised model.
- 3) To recommend a ranking of houses on the basis of user preferences, which will later be evaluated in order to improve the quality and relevance of the recommendations.

The remainder of this document is organized as follows: Section II provides a review of the available literature on fuzzy housing recommendation systems, while Section III presents the proposed method along with details of the algorithms employed. Section IV shows an analysis of the dataset used for experimental evaluations, and Subsections IV-C and IV-E present a description of the proposed algorithms for dimensionality reduction and record clustering, leading to the most recommended houses. Section V presents a comparative analysis of the proposed technique and the state of the art. Finally, the paper concludes in Section VI by highlighting various aspects of the proposed approach and possible future directions.

II. RELATED WORK

A fuzzy recommendation system for housing selection integrates multiple-criteria decision-making methods with which to help low-income individuals choose suitable housing options [14]. Moreover, the application of extensive data profiles and artificial intelligence to the housing provident fund system enhances digital management and service levels, promoting intelligent government construction [14], [15]. Evaluating customer preferences regarding housing location using Fuzzy Full Consistency and Fuzzy Measurement Alternatives methods helps determine the most suitable district in which to purchase a house, with price being a crucial criterion [16]. Moreover, a multi-criteria recommendation system (MCRS) based on user ratings and similarity methods provides housing recommendations with varying levels of accuracy, such as 88.7% in the case of the Pearson correlation [17]. These systems leverage data analysis and decision-making techniques so as to streamline housing selection and improve user satisfaction.

Table 1 summarizes relevant information extracted from various papers, including the reference, the technique used, the data analyzed, and the key findings, which help and contribute to this research.

A. DATA SOURCES

The Fuzzy Recommender system for Housing utilizes various data sources to recommend housing selection. It leverages big data from the Housing Provident Fund Center to create a multidimensional profile model and establish a housing

TABLE 1. Table of relevant information obtained from papers.

Reference	Technique	Data	Findings
[18]	Model 2SA-RERec (Two Sentiment Analysis of Real Estate Recommendation)	The study uses economic and sentiment data to forecast real estate sales and evaluate the accuracy of the proposed model.	Based on sentiment analysis and economic data in order to enhance ad recommendations.
[19]	Fuzzy trapezoidal critical path method	Project data was obtained from the housing project.	The shape of the fuzzy diagram affects the total duration range of the project.
[20]	Fuzzy Logic Algorithm and Analytic Network Process (ANP)	Munawar Kos and Diding Kos are the best-recommended boarding houses.	Munawar Kos had the highest value, 6.55%. Deciding Kos followed with a value of 6.52%.
[16]	Fuzzy FUCOM And Fuzzy MARCOS Approaches	Yakutiye had the highest number of home sales between 2015 and 2021.	Price is the most essential criterion as regards selecting a house.
[22]	Various applications of Artificial Intelligence	Several criteria are taken into account in order to determine the location of the housing.	Entrepreneurs save time and reduce risks in housing development.
[23]	Application of intelligence	Implementation of sensors in order to manage a healthy environment.	Implementing intelligent technologies can optimize the quality of life for residents in housing.
[24]	PenerapanFuzzy SAW	Data was collected from a research study in the Sepakek, Lombok Tengah village.	The system provides accurate recommendations with which to assist in the construction of non-livable housing.
[15]	Two-Tower Neural Collaborative Filtering	Artificial intelligence recommendation model based on previous purchase transaction history.	A housing recommendation model built using a neural network
[14]	System Using Multi-Criteria Decision-Making Methods	Housing selection criteria for low-income individuals.	Integrated Best Worst (BW) and TOPSIS methods for decision-making.
[17]	Known Ratings-Based Multi-Criteria	User ratings of 14 criteria used for recommendations.	The Pearson correlation method attains the highest accuracy as regards recommendations.
[28]	System Using Hybrid User-Centric, System Using Fuzzy Logic and Item-Based Collaborative Filtering	Local area lease house.	The goal is to assist homeowners, relocated populations, and land agencies.
[29]	SeoulHouse2Vec": An embedding-based collaborative filtering housing recommender system for analyzing housing preference	This system leverages public big data and GIS data so as to expand the search for housing alternatives across the city and ultimately facilitate decision-making for various stakeholders, such as policymakers and urban planners.	SeoulHouse2Vec is an innovative housing recommendation system designed to analyze individual housing preferences in Seoul. Embedding-based collaborative filtering maps users and housing options onto a dense vector space, enabling personalized recommendations that reflect atypical preferences.
[30]	Improving recommender systems via a dual training error based correction approach. Expert Systems with Applications	The method is evaluated using four real-world datasets in order to demonstrate its effectiveness.	Improves the accuracy of recommendations in recommendation systems using the Dual Error Correction Method (DTEC).

TABLE 1. (Continued.) Table of relevant information obtained from papers.

[31]	An effective distributed predictive model with Matrix factorization and random forest for Big Data recommendation systems.	Proposes a distributed solution for recommendation systems in Big Data that enhances prediction quality and reduces computational cost using Apache Spark.	The proposal is evaluated by means of experiments carried out on three real datasets, demonstrating improvements as regards mean absolute error (MAE), root mean square error (RMSE), and computation time when compared to existing methods.
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profile tagging system [28]. The system additionally incorporates the variables of roof, floor and wall as input data for fuzzy logic methods with which to classify housing conditions [15]. Moreover, the system uses fuzzy database models to calculate membership grades and fire intensity in order to make housing recommendations based on consumer criteria [29]. The integration of these diverse data sources allows the Fuzzy Recommender system for Housing to streamline the decision-making process such that consumers and developers can select appropriate housing options.

B. ALGORITHMS

The Housing Fuzzy Recommender system utilizes the Analytic Network Process (ANP) method for calculations and Fuzzy Logic for criteria weighting [20]. The system additionally incorporates recommendation methods based on extensive data profiles and artificial intelligence, specifically a two-tower neural collaborative filtering algorithm, to enhance digital management and provide housing recommendations based on the previous purchase transaction history [15]. Conversely, a study evaluating a shared apartment recommendation system in metropolitan areas found that several algorithms had been implemented, leading to performance disparities among different population groups and affecting rental opportunities for certain marginalized [30]. Furthermore, a proposed hybrid recommendation system has been designed with the user in mind. This integrates the associative exploration approach of Fuzzy Neural Networks (Anfis) with collaborative refinement in order to provide personalized recommendations based on user preferences and item popularity [31]. This user-centric approach could revolutionize the housing recommendation process, ensuring that individual needs are met.

C. USER PREFERENCES

The Fuzzy Recommender system for Housing, a unique platform that leverages fuzzy logic and item-based collaborative filtering, has been designed to cater for the needs of potential home buyers and renters. It stands out from other systems by managing user preferences and generating lists of houses on the basis of these preferences [32]. This preference-based approach has, as opposed to using defined rewards, been proven to enhance long-term user engagement in recommendations. The system introduces PreRec, an innovative concept that employs user preferences

to train a reward function, thus surpassing previous methods as regards optimizing long-term engagement [33]. By incorporating fuzzy logic, the system can adapt to varying levels of user interest over time, ensuring that user preferences are accurately captured and reflected in the housing options recommended [34]. The study proposes a fuzzy inference system based on user preferences in order to recommend images on social networks. It uses a method that extracts and selects features from preferred images, thereby improving accuracy and recall when compared to other approaches [35].

D. CHARACTERISTICS OF RECOMMENDATION SYSTEMS

The model has a multi-criteria approach with which to select housing for low-income individuals, combining best-worst (BW) and TOPSIS methods in order to identify critical criteria and recommend homes that best fit their needs and desires [14]. The main feature of the model is its focus on developing a decision support system in the real estate sector, using the Fuzzy Tsukamoto method to recommend homes on the basis of criteria such as price, location, quality of construction, and other relevant features for consumers [36]. The main feature of this model is its focus on creating a user-centered hybrid housing recommendation system, using fuzzy logic and item-based collaborative filtering to generate customized lists of houses according to the individual preferences of potential buyers and tenants. It also offers a virtual tour of homes in order to facilitate decision-making [37]. Its focus is on using software programs based on fuzzy artificial intelligence to help real estate developers optimize the location of their construction projects. These programs use decision criteria determined by the entrepreneur to reduce the risks and time associated with selecting housing locations [21]. The Fuzzy Recommender system for Housing stands out owing to its content-based filtering approach designed for search engines, and efficiently manages large amounts of user and housing data for recommendations [38]. It is distinguished by the fact that it employs user spending behavior, geographic data segmentation, local scoring, and personalized housing recommendations based on these factors [39].

Fuzzy recommendation systems for the real estate sector have recently been studied owing to their ability to manage market imprecision. A systematic review of 1,003 papers published between 1985 and 2024 was conducted using the PRISMA 2020 methodology. The analysis showed that

65% of algorithms employ fuzzy logic, and 80% assess performance. The main challenges include integrating diverse sources of information. The studies reviewed highlight the need to incorporate contextual information and explore hybrid approaches in order to improve the performance of these systems [40].

III. PROPOSED HOUSING RECOMMENDATION METHODOLOGY

This section presents the proposed methodology with which to collect information, model, and recommend suitable housing. This methodology covers the steps required in order to identify user preferences, housing characteristics and critical elements. The proposed model is based on a series of detailed steps, which are illustrated in Figure 1.

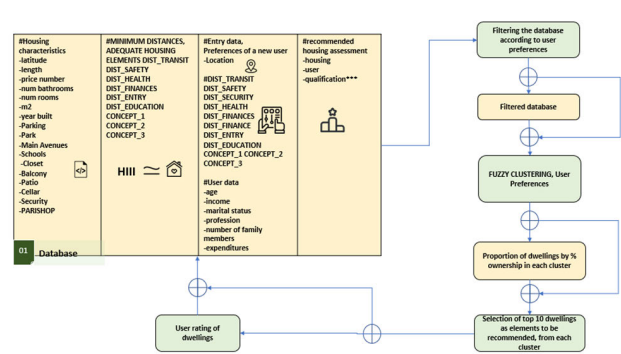


FIGURE 1. Procedure employed for housing recommendation.

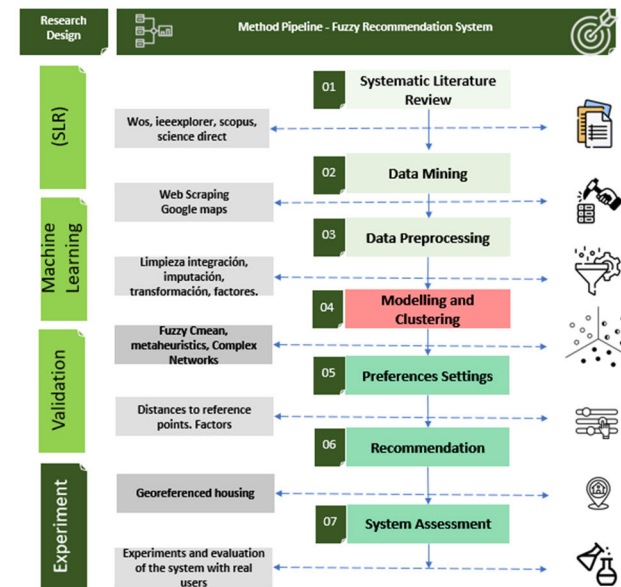


FIGURE 2. Research design and method pipeline.

The research design for the fuzzy housing recommendation system presented in Figure 2 is organized into several stages: it begins with a literature review carried out in order to identify relevant algorithms and techniques, followed by the

extraction and preprocessing of real estate and spatial data through the use of web scraping. A Fuzzy C-Means model is then applied so as to capture users’ latent preferences, which are validated by a user simulator agent. Finally, the system is evaluated with an MVC architecture on a web platform on which real users state their preferences, receive recommendations and rate their satisfaction, thus providing system performance metrics. The housing recommendation system initially employs content-based filtering, generating recommendations according to the users’ initial preferences. As ratings are collected over time, a collaborative filtering method is integrated in order to suggest properties that are highly rated by users with similar profiles. In cases of limited data, a hybrid approach is used, which combines both methods so as to enhance accuracy. This recommendation filtering flow is presented in Figure 3 and dynamically is adjusted in order to maximize the relevance of recommendations, thus optimizing the users’ experience.

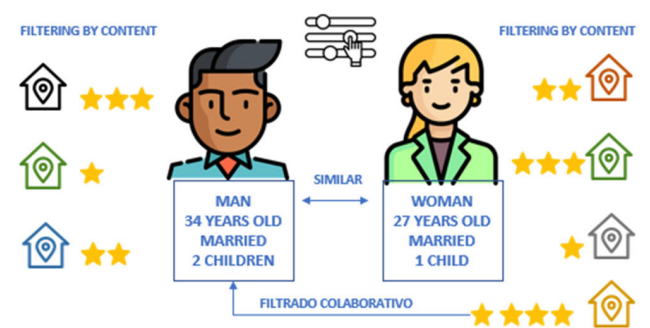


FIGURE 3. Hybrid filtering.

The MVC pattern is very popular in web and desktop application development. It facilitates the separation of responsibilities, thus enabling a more structured development process with less dependency among business logic, presentation and user interactions.

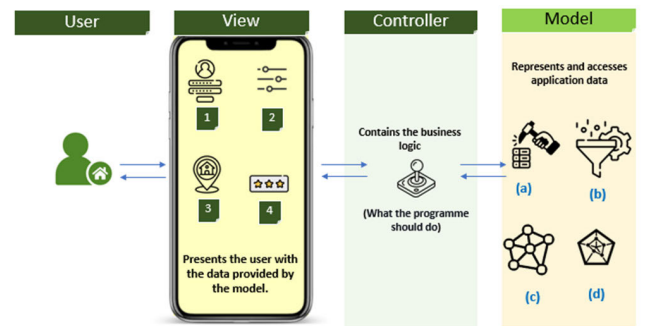


FIGURE 4. Recommender system architecture.

In the MVC flow presented in Figure 4, the user interacts with the view, after which the controller receives that input, processes it and updates the model as necessary. The model then updates the data, and the view reflects the changes made by displaying the new information to the user.

A. ALGORITHM

- 1) Data Collection:
Use web scraping to gather housing characteristics from real estate websites.
Collect geospatial data points of the houses and critical locations for suitable housing.
- 2) Spatial Distance Calculation:
Use QGIS to calculate spatial distances from the houses to critical locations.
- 3) Dimensionality Reduction:
Apply factor analysis in order to reduce the dimensions of the housing characteristics.
- 4) Variable Transformation:
Normalize the variables using the formula:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (1)$$

- 5) Logarithmic Transformation:
Apply the base-10 logarithmic transformation, adding 1 in order to avoid indeterminacy:

$$\log_{10}(x_i + 1) \quad (2)$$

- 6) Unsupervised Modeling:
Use Jambu's elbow method or genetic algorithms to determine the number of clusters.
- 7) Model Fitting:
Test different clustering algorithms such as k-means, fuzzy c-means, DBSCAN and HDBSCAN.
- 8) Dimensionality Reduction for Visualization:
Apply the PCA (Principal Component Analysis) in order to visualize the number of clusters.
- 9) User Preference Selection:
Allow the user to select preferences for location and key housing elements on a scale from 0 to 100%.
- 10) Cluster Membership Model:
Use the Fuzzy C-means algorithm to assign a membership range to each cluster.
- 11) Housing selection using the Bat metaheuristic algorithm: Selects a percentage of properties from each cluster using Bat Optimization. Top Housing Recommendations:
Recommend a top selection of houses on the basis of the user's selection criteria.
- 12) Similarity Calculation:
Using cosine distance, calculate a similarity value in order to validate the model between user preferences and houses.
- 13) User Rating:
Allow the user to rate each house recommended by the web application.
- 14) Recommender Memory:
Store this search in graphs, in which each preference is a node. Each node contains the recommended houses, similarities and ratings so as to optimize computation time for future similar preferences.

Our data analysis process is powered by Python and a range of related tools. Each step on our journey, from data collection to analysis, requires specific code in Python. We also utilize tools such as Selenium for web scraping, QGIS for spatial distance calculations, and libraries such as scikit-learn for the factor analysis, normalization, clustering and dimensionality reduction. These tools, which are combined with Python, form the backbone of our data analysis process.

The process starts with data collection via web scraping in order to obtain detailed housing characteristics from multiple online sources and relevant geospatial data such as the locations of nearby services, parks, schools, etc. The spatial distances from each house to these key points are then calculated using GIS tools such as QGIS, thus providing a comprehensive perspective of the surroundings of each property.

Dimensionality reduction is then performed using factor analysis, which helps identify underlying patterns in the data and reduce complexity. Housing characteristics are subsequently normalized in order to ensure that all the features have an equal weight in the analysis. A logarithmic transformation is additionally applied so as to smooth out the effects of extreme values and improve the stability of the model.

Unsupervised clustering techniques such as k-means and fuzzy c-means are then used in order to group houses into homogeneous clusters based on their characteristics and geographic locations. This is a crucial step, as it allows the identification of similarity patterns among the properties. These patterns are then used to power a recommendation system, which facilitates the recommendation of similar houses to users.

Users can specify their preferences by means of a location, size, price and service proximity rating system. The Fuzzy C-means algorithm is employed in order to assign degrees of membership to each house in each cluster, reflecting the flexibility and uncertainty in the group memberships.

Houses are then recommended using the nearest neighbor method within each cluster, prioritizing those closest to the cluster centroid and best matching the user's preferences. Cosine distance is used to calculate the similarity between user preferences and the characteristics of the recommended houses, thus validating the suitability of the recommendations.

Finally, the user has the opportunity to rate and provide feedback on the recommended houses. This feedback is crucial, as it allows the system to continuously adjust and improve the recommendations in future iterations. This commitment to accuracy ensures an increasingly personalized and accurate recommendation experience, thus instilling confidence in the user.

B. FUZZY HOUSING SEGMENTATION

Factor Analysis: (Applied to housing characteristics). A factor analysis is a statistical technique that seeks to explain the variability of a set of observed variables in terms of a potentially smaller number of unobserved variables called

factors [41]. There is no single formula, but the technique generally involves the following matrix transformation:

$$X = \Lambda F + \epsilon \quad (3)$$

where:

X is the matrix of observed data (with dimensions $n \cdot p$, where n is the number of observations and p is the number of variables).

F is the matrix of latent factors (with dimensions $n \cdot k$, where k is the number of factors).

L is the matrix of factor loadings (with dimensions $k \cdot p$).

ϵ is the matrix of errors (with dimensions $n \cdot p$).

A factor analysis aims to estimate matrices F and L , which explain the most variability in X with the fewest possible factors.

The Fuzzy C-means algorithm assigns degrees of membership to each cluster point rather than strictly assigning each end to a single cluster [42]. The formula employed in order to calculate the degree of membership of point i to cluster j is:

$$u_{ij} = \frac{1}{\sum_{k=1}^c \left(\frac{d_{ij}}{d_{ik}} \right)^{\frac{2}{m-1}}} \quad (4)$$

where:

u_{ij} is the degree of membership of point i to cluster j .

x_i is the point i in the feature space.

v_j is the centroid of cluster j .

c is the number of clusters.

m is the membership weighting parameter (typically set at 2)

C. HOUSING RECOMMENDATION

H. A Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms a set of correlated variables into a new set of uncorrelated variables called principal components [43]. The formula used in order to calculate the principal components involves computing the eigenvectors of the covariance matrix of the data (applied in order to represent clusters in 2D):

1. Compute the covariance matrix C of the data.
2. Compute the eigenvectors and eigenvalues of C .
3. Sort the eigenvalues in descending order and select the k eigenvectors corresponding to the k largest eigenvalues.
4. Project the original data onto the subspace defined by the chosen k eigenvectors.

The k-Nearest Neighbors (kNN) algorithm is a supervised learning method for classification and regression. In order to classify a new data point, the kNN algorithm searches for the k nearest points in the training set and predicts the label of the new point on the basis of the labels of its neighboring points. The formula used in order to calculate the distance between two points, A and B, may vary, but the Euclidean distance is

commonly used:

$$d_{E(x,y)} = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (5)$$

Cosine Similarity: (Applied in order to calculate the similarity between user preferences and recommended houses).

Cosine distance: measures the angle between two vectors in a multidimensional space. It is used to measure the similarity between two vectors, regardless of their magnitude [44]. The formula employed in order to calculate the cosine distance between two vectors A and B is:

$$d_{\text{coseno}(A,B)} = 1 - \frac{A \cdot B}{|A||B|} \quad (6)$$

$A \cdot B$ is the dot product of vectors $|A|$ y $|B|$, which are the Euclidean norms of vectors A and B, respectively.

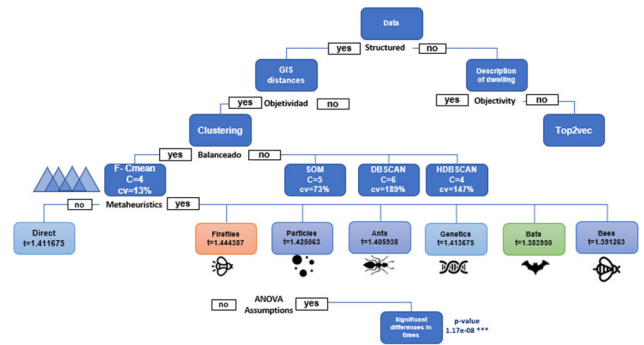


FIGURE 5. Pipeline of the method used in the recommender system.

Figure 5 illustrates the pipeline for a recommendation system that combines clustering techniques and bio-inspired algorithms. The process begins with data input, which is classified into structured and unstructured data. For structured data, GIS distances are calculated, while unstructured data is analyzed using Top2Vec, although this method may be subjective owing to the housing descriptions provided by sellers. Methods such as F-Cmean, SOM, DBSCAN, and HDBSCAN are employed to improve the homogeneity of clustering, along with metaheuristics when necessary. Finally, an ANOVA analysis evaluates significant differences in execution times, thus ensuring that the results are statistically significant. This approach optimizes the data's spatial and descriptive characteristics in the recommendation system.

IV. EXPERIMENTAL EVALUATION

A. DATASET

The distances to critical elements of housing play a fundamental role in evaluating the location and convenience of a property. Some of these elements are ENTERTAINMENT facilities such as parks and gyms, totaling 216 locations; HEALTH, represented by hospitals and clinics, totaling 56 locations; TRANSPORT, including bus stops and taxi stands, totaling 61 locations; SECURITY with surveillance

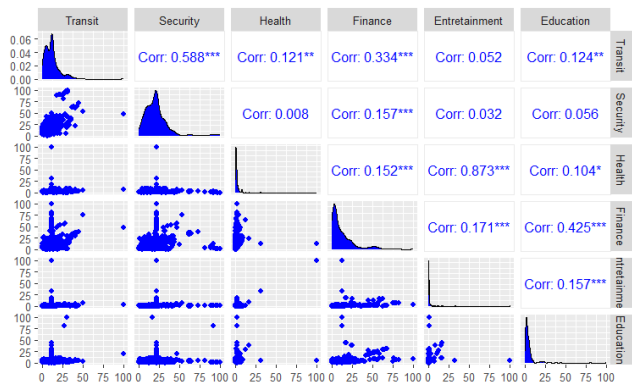


FIGURE 6. Exploratory Data Analysis: distances of elements.

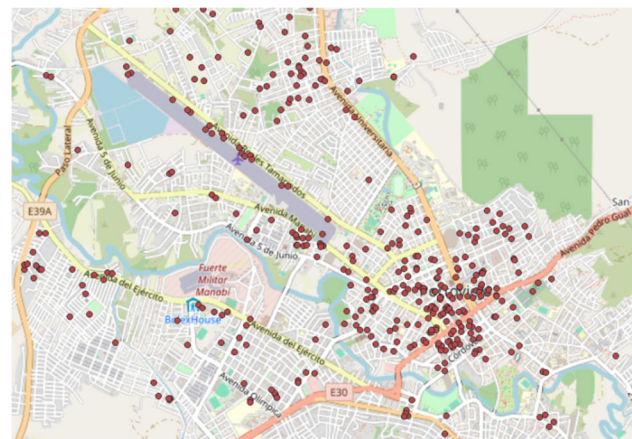


FIGURE 7. Spatial data on dwellings.

cameras and nearby police units, totaling 41 locations; EDUCATION covering everything from daycare centers to universities, totaling 171 locations; COMMERCE including markets and squares, totaling 284 locations, and FINANCE, which is represented mainly by the presence of banks, totaling 103 locations. These scores reflect the quantity of each type of location near the housing, significantly impacting residents' quality of life and convenience.

Figure 6, which shows, the Exploratory Data Analysis (EDA) provides a comprehensive visual representation of the relationships between the variables of element distances. The scatter matrix illustrates how each variable relates to the others, enabling the identification of patterns and correlations among them. The histograms additionally provide a detailed view of the individual distribution of each variable, highlighting the frequency of observed values and potential trends. This combination of tools allows an in-depth understanding of the data structure and is essential as regards carrying out an informed data analysis and making modeling decisions.

B. PRE-PROCESSING OF THE DATASET

Figure 7 shows the spatial data regarding the dwellings represented on a map, which visualizes the geo-referenced locations of the properties in Portoviejo.

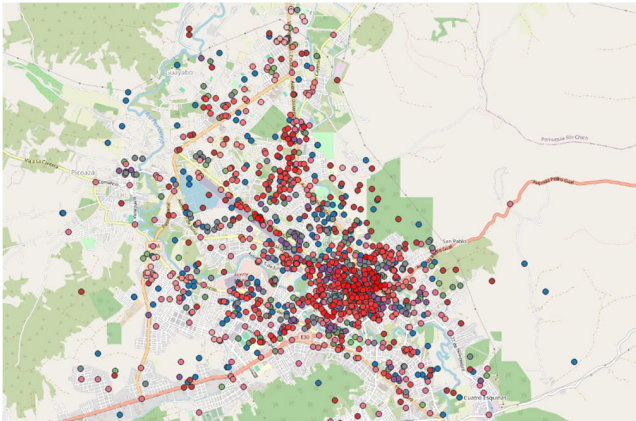


FIGURE 8. Spatial data of key elements.

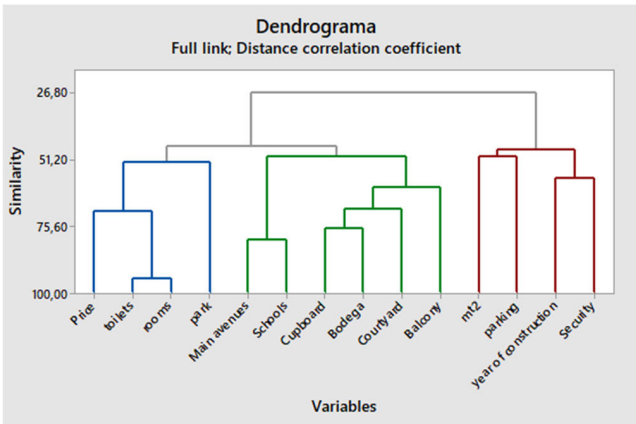


FIGURE 9. Dendrogram employed to organize the factors of housing variables.

Figure 7 of the analysis presents spatial data on housing, while Figure 8 highlights critical elements related to health, education, commerce, security, and finance. This map shows the geographic locations of houses in one layer, while separate layers represent points corresponding to essential elements such as hospitals, schools, supermarkets, police stations, and banks. This spatial representation provides a comprehensive view of the housing environment, facilitating the identification of proximity to essential services and points of interest for potential residents. This geospatial analysis is fundamental as regards understanding the accessibility and convenience of homes with regard to nearby services and facilities, which may significantly influence location and property purchasing decisions.

Figure 9, a dendrogram, presents a hierarchical visualization of the organization of factors derived from housing variables. This dendrogram graph shows the clustering structure of the factors, in which longer branches and deeper divisions represent greater distances between groups of variables. This representation makes it possible to identify clustering patterns and relationships among different housing

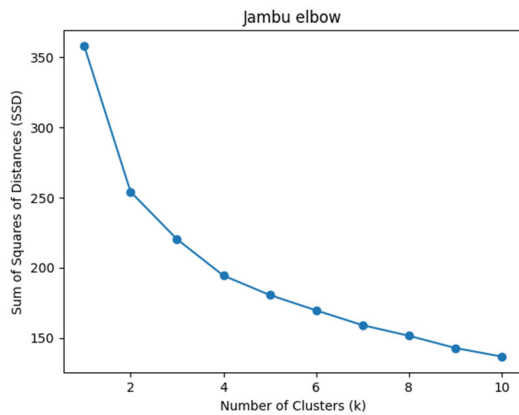


FIGURE 10. The optimal number of clusters recommended by Jambu's elbow method.

variables, thus facilitating an understanding of the underlying data structure. The dendrogram is a valuable tool with which to explore the complexity of relationships between variables, and can provide essential insights that enable the data to be analyzed and interpreted more efficiently.

The groups of variables are the concepts, factors or latent variables. (dendrogram and factor analysis) A base ten logarithm transformation is applied to normalized values if this makes sense in a particular data and analysis context. Logarithmic transformation can help reduce asymmetry in data distribution or handle values spanning extensive ranges. When applying base ten logarithms to values equal to 0, the result will be -infinity because the logarithm of 0 is undefined. One is, therefore, added to the data so as to avoid this issue before applying the logarithm. This is known as "smoothing" the data.

C. MODELING

Figure 10 shows the optimal number of cluster analyses when using Jambu's elbow method. This graph depicts the number of clusters on the x-axis and the intra-cluster dispersion measure on the y-axis. A curve shows how intra-cluster dispersion decreases as the clusters increase, but with a less steep slope loss after a certain point. The application of the elbow criterion makes it possible to identify the inflexion point at which the slope loss is less pronounced, suggesting that the optimal number of clusters is 4. This figure provides a helpful guide with which to select the appropriate number of clusters in a clustering analysis, facilitating the interpretation and effective segmentation of data.

Figure 5 illustrates the process of determining the optimal number of clusters using Jambu's elbow method. Although metaheuristic algorithms such as genetic and particle algorithms were applied, they yielded suboptimal results, with only two completely unbalanced clusters. Moreover, HDBSCAN algorithms automatically provided up to 33 clusters, which was impractical, and Jambu's elbow method was therefore recommended, as it yielded

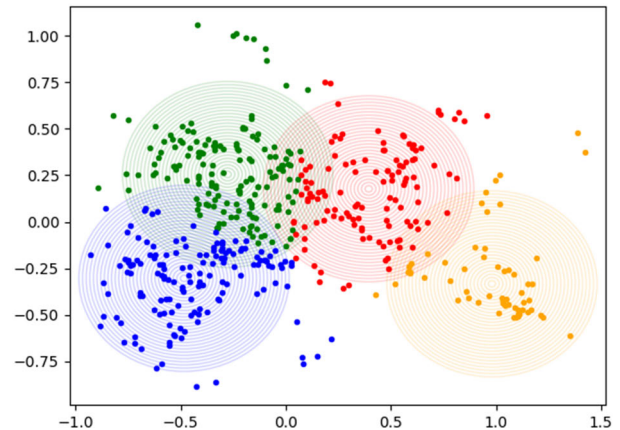


FIGURE 11. Dimensionality reduction of data using PCA to represent the fuzzy clusters on a 2D plane.

an optimal number of clusters, in this case, four, by identifying where the slope loss was less pronounced. This selection was considered most suitable for further analysis, since it allowed a more balanced and meaningful data segmentation.

Figure 11 depicts data dimensionality reduction using a Principal Component Analysis (PCA) to visualize the clusters on a 2D plane. This graph shows how the original features of houses are transformed into two principal components that capture most of the data variability. Each point on the 2D plane represents a house, and its position reflects its distribution in the reduced feature space. The points are colored according to their cluster membership, allowing a visual identification of the separation and distribution of clusters in the two-dimensional space. This simplified representation facilitates the interpretation of the data structure. It provides a clear visualization of how houses are clustered on the basis of their characteristics, which can be helpful for further analysis and decision-making.

Figure 12 shows the dimensionality reduction of data using a Principal Component Analysis (PCA) to represent the clusters on a three-dimensional (3D) plane. In this graph, each point represents a house in the three-dimensional space, in which the three principal dimensions capture most of the data variability. The points are colored according to their cluster membership, thus allowing the easy identification of the distribution of clusters in the three-dimensional space. This three-dimensional representation provides a more complete and detailed visualization of the data structure when compared to the 2D plane representation, enabling a better understanding of the separation and distribution of the groups. This visualization is useful for exploratory data analysis and can provide valuable insights for decision-making in subsequent studies.

D. TRAINING AND EVALUATING

Figure 13 shows the control panel that the users can employ in order to select their preferences. This panel comprises

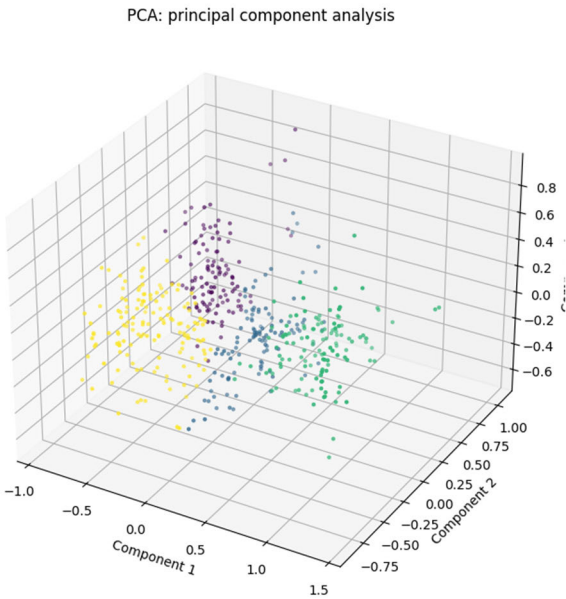


FIGURE 12. Dimensionality reduction of data using PCA to represent the clusters on a 3D plane.

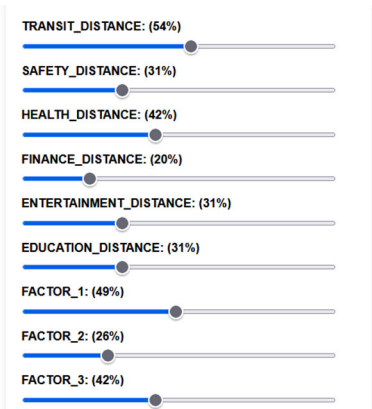


FIGURE 13. Control for selecting user preferences.

six sliders representing distances to critical elements such as healthcare, education, commerce, security and finance. These sliders allow users to adjust their preferences on the basis of the relative importance of proximity to these essential elements, ranging from 0% (indicating that proximity is not necessary) to 100% (indicating maximum importance). There are also three additional sliders corresponding to factors representing house characteristics such as size, price and age, which similarly permit the users to adjust the relative importance of these features. This control panel provides an intuitive interface that allows users to personalize their preferences and reflect their needs when searching for suitable housing.

The computational complexity analysis of the various metaheuristics shown in Table 2 reveals that, although the Bat method was the fastest on average, all iterative algorithms have a complexity of $O(n \cdot \text{num_iterations} \cdot \text{population})$,

TABLE 2. Complexity analysis.

Algorithm	Execution Time Average (s)	Complexity
Firefly	1,444387	$O(n \cdot \text{iterations} \cdot \text{fireflies})$
PSO	1,425063	$O(n \cdot \text{iterations} \cdot \text{particles})$
GA	1,413675	$O(n \cdot \text{generations} \cdot \text{population})$
Direct	1,411675	$O(n)$
Ant Colony	1,405938	$O(n \cdot \text{iterations} \cdot \text{ants})$
Bee	1,391263	$O(n \cdot \text{iterations} \cdot \text{bees})$
Bat	1,38395	$O(n \cdot \text{iterations} \cdot \text{bats})$

where ‘population’ refers to the number of individuals in each algorithm, such as particles, ants, or bats. Each technique varies in approach, from position updates in PSO to distance calculations in Fireflies and route selection in Ant Colony. It involves additional costs based on the interactions and operations required in each iteration. The Direct method has the lowest complexity, $O(n)$, as it does not require iterations or additional operators.

E. APPLICATION

Figure 14 shows a web application interface designed in order to test the usefulness of the housing recommendation model and allow users to rate the recommendations received. The web application has an intuitive interface with which users can visualize recommended homes, along with their key features and locations, on an interactive map. Each recommended home is accompanied by detailed information, such as size, price, distance to essential amenities and other relevant factors. Users can rate each home on the basis of their satisfaction, providing valuable feedback to the model. In addition, the web application can offer additional functionalities, such as the ability to adjust user preferences, browse other recommended homes and receive updated recommendations in real-time. Overall, this web application provides an effective platform with which to evaluate and improve the housing recommendation model and meet the individual needs and preferences of users in their search for suitable housing.



FIGURE 14. Web application employed to test the usability of the model and rate housing recommendations.

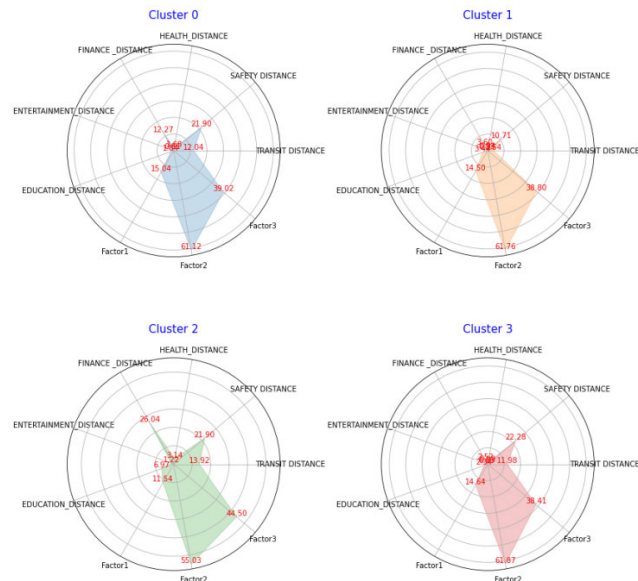


FIGURE 15. Radar plot of the values of the clusters according to the variables of the model.

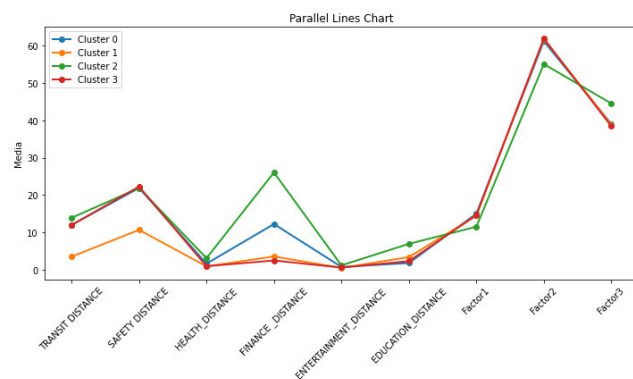


FIGURE 16. A parallel line graph is used to compare the clusters' values according to the variables of the model.

F. VALIDATION

Simulation of user preferences and calculation of cosine similarity in the recommendations.

Figure 15 illustrates each cluster is represented as a polygon on the graph, with each point on the line representing the value of a specific variable. This makes it easy to compare cluster profiles and understand how they differ according to the variables of the model, which is helpful as regards analyzing patterns and trends in the segmented data.

Figure 16 displays a graph. Each vertical axis represents a variable, with parallel lines connecting the corresponding values of each variable for each group or cluster. This effective visualization method allows the easy identification of differences and similarities in variable values among clusters, helping to identify patterns and trends in the segmented data.

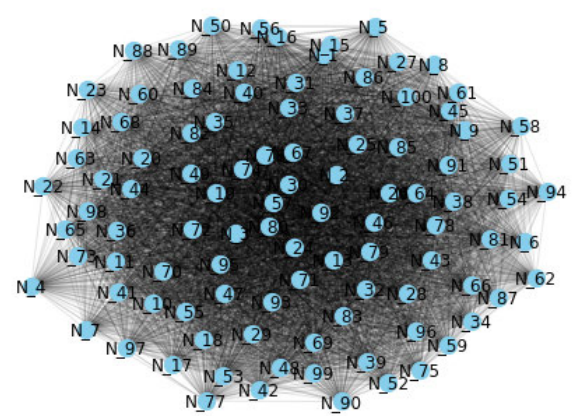


FIGURE 17. Memory of searches according to simulated user preferences, represented in graphs.

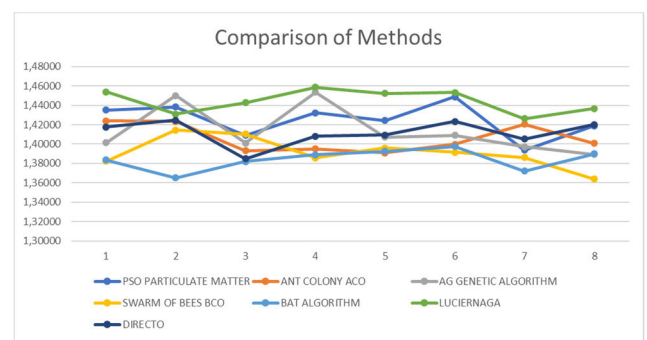


FIGURE 18. Comparison of trends of metaheuristics according to runtime.

The distance of each group of dwellings from each cluster is calculated, after which the similarity is calculated.

Figure 17 shows the memory of the graph-based housing recommendation system, in which each node represents a housing preference with attributes such as distances and ratings, and the edges indicate similarities among them, thereby creating a memory that facilitates recommendation search. When a new preference is introduced, the system calculates its similarity with existing nodes and selects the most similar node if it exceeds a threshold. Otherwise, it employs a fuzzy model developed in the research presented herein. This approach enhances the accuracy and speed of recommendations, with a search time of 0.0009593963623046875.

Figure 18 compares the trends of different metaheuristics on the basis of execution time, showing the times (in seconds) on the Y-axis and various experiments or iterations on the X-axis. The time variability ranges from 1.32 to 1.48, with algorithms such as PSO having significant fluctuations (1.35 to 1.42), while ACO and AG have a more consistent performance (ACO: 1.38 to 1.44). BCO has high variability, especially in experiment 4, and Bat remains stable (1.38 to 1.42). Firefly is the slowest, reaching 1.46, and Direct has moderate variability. In conclusion, PSO and BCO are suitable for speed, while Bat and Direct are preferable for stability.

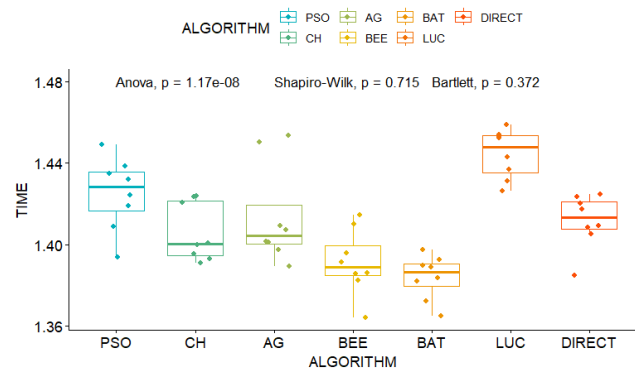


FIGURE 19. Comparison of execution times of metaheuristics: ANOVA and assumption verification.

TABLE 3. ANOVA used in order to evaluate statistically significant differences in metaheuristic times.

Summary of the ANOVA					
ALGORITHM	df	Sum Sq	Mean Sq	F-value	Pr(>F)
Residuals	6	0.01973	0.003289	12.85	1.17e-08***
	49	0.01254	0.000256		

Figure 19 comprises a boxplot that compares the dispersion and median execution time (Y-axis) of several algorithms: AG, BAT, BEE, CH, DIRECT, LUC, and PSO (X-axis). Each boxplot shows the median, interquartile range, extreme values, and outliers. AG has a median of approximately 1.42 and has outliers. BAT is faster, with a median below 1.40 and low dispersion. BEE stands out owing to its low median (1.38) and consistency. CH and DIRECT have similar medians (1.41), but with less dispersion. LUC is the slowest, with a median of 1.45 and high variability. In conclusion, BEE and BAT are rapid and consistent options, while LUC may not be ideal if execution time is prioritized.

Table 3 presents an analysis of variance (ANOVA) used in order to assess statistically significant differences in execution times among various metaheuristics. A summary of the model (md1) is shown with 6 degrees of freedom for ‘ALGORITHM’ and 49 for ‘Residuals.’ The sum of squares for ‘ALGORITHM’ is 0.01973, while that for ‘Residuals’ is 0.01254. The F value is 12.85, indicating the relationship between the variability explained by the model and that which is unexplained, while the p-value of 1.17e-08 indicates that the differences are highly significant, suggesting that at least one of the metaheuristics has a significantly different execution time from the others.

Figure 20 presents multiple comparisons using Tukey’s test on the times utilized by various metaheuristics. The group column uses letters to group the metaheuristics according to their performance, where identical letters indicate that the means of the times in those groups are not significantly different. For example, ‘Firefly’ belongs to group ‘a,’ indicating an inferior performance, while ‘Particle’ and ‘Genetic Algorithm’ are in group ‘b,’ indicating a similar performance. Likewise, ‘Ant Colony’ and ‘Bees’ are grouped in ‘cd’ and

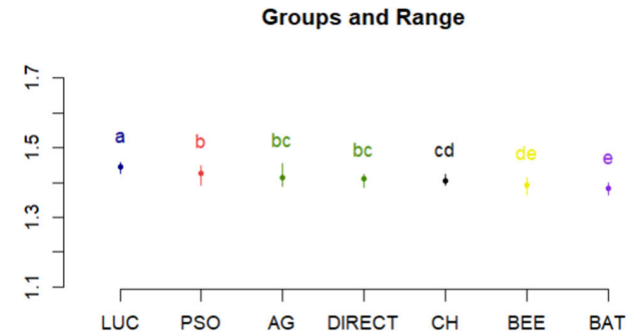


FIGURE 20. Tukey’s multiple comparisons of the times used by the metaheuristics.

‘de,’ respectively, suggesting that their execution times are statistically comparable.

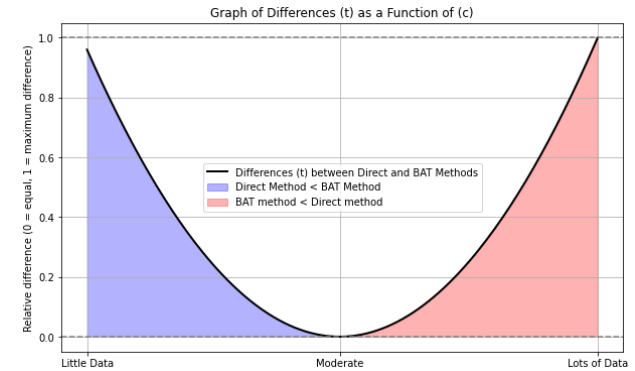


FIGURE 21. Time difference between a direct process and a metaheuristic method depending on the amount and complexity of the database.

Figure 21 illustrates the difference in times between a direct process and a metaheuristic method based on the quantity and complexity of the database. The times (t) are represented by the amount of data (c), showing how the use of metaheuristic methods can significantly optimize processing time when compared to direct approaches.

G. VALIDATION OF THE RECOMMENDATION MODEL

Figure 22 displays the performance of a recommendation model across four key metrics: precision, recall, diversity and user satisfaction. The model has a high precision of 0.816, indicating that most recommendations are relevant. The recall, with a value of 0.89, suggests that the model retrieves the most relevant items. Diversity, with a value of 0.175, shows that there is an adequate variety in the recommendations provided. However, user satisfaction was 2.9 out of 5 in the experiment with real users, equating to 0.58 positive recommendations. This suggests that there is room for improvement as regards the perception of the recommendations. Using more data and conducting additional testing is recommended so as to fine-tune and optimize the



FIGURE 22. Metrics used in order to validate recommender system performance.

model. The graph provides a clear view of the performance of the model, highlighting its strengths and the areas that need improvement.

V. COMPARATIVE ANALYSIS

We now show the comparative analysis of the results obtained, which was carried out with the aim of improving the accuracy and authenticity of our results.

The radar plot shows the performance of a specific recommendation model on four key metrics: precision, recall, diversity and user satisfaction. In this model, there is high precision (0.816), indicating that most of the recommendations are relevant, and moderately high recall (0.89), suggesting that the most appropriate items are retrieved. Diversity is moderate (0.175), reflecting a reasonable variety in recommendations, while user satisfaction achieved a score of (0.58). This score could be improved by means of additional testing with collaborative filtering, optimizing the model on the basis of honest user feedback. This plot provides a visual representation of the performance of the model as regards multiple essential aspects, thus facilitating the identification of its strengths and those areas that need improvement.

Diversity in recommender systems is crucial as regards improving the user experience and preventing information bubbles. Traditional recommendation models often focus solely on accuracy, leading to repetitive suggestions based on past preferences [45], [46]. In order to address this limitation, it is essential to incorporate diversity so as to recommend specialized articles and maintain user interest [45]. Several approaches with which to balance accuracy and diversity have been proposed, such as using deep latent factor models with a diversity cost/penalty ratio to control the balance effectively [47].

Variability in the results of metaheuristic methods is a common characteristic due to their stochastic nature. This variability can be managed by using initial seeds, which allow specific conditions to be replicated across multiple

runs. By controlling the seeds, uncertainty in execution times and results can be reduced, facilitating the comparison and analysis of experiments. As described in the article [48], the strategic insertion of predefined candidate solutions into the optimization process can stabilize the algorithm’s performance, improving consistency and efficiency in searching for solutions.

Recommender systems influence user satisfaction, as several factors affect the overall experience. Research suggests that measuring user satisfaction, combating the scarcity of satisfaction signals and adapting the training of recommenders are crucial to improving user journeys [49], [50]. Factors such as presentation modality and explanation content also significantly influence user satisfaction with recommender systems, with different effects observed for text and voice explanations [50]. Furthermore, users’ attitudes towards personalized recommendations can impact digital satisfaction, and a positive correlation has been found between attitudes toward personalization and overall satisfaction, highlighting the importance of understanding users’ preferences and experiences in recommender systems [51]. By addressing these aspects, recommender systems can be optimized so as to improve user satisfaction and provide more personalized and engaging experiences.

Precision and recall are crucial metrics with which to evaluate the performance of recommender systems. Several studies highlight the importance of balancing precision and diversity in recommendations [52].

TABLE 4. Complexity and failure analysis.

ComplexityArea	Description	Representation
Pre-processing	Use of Selenium and georeferencing with Google Maps	Slowextraction
Modeling	Algorithms such as Fuzzy C-Means and metaheuristics	Computationally intensive
SystemFailures	Lackof data orcoldstart	Limited prediction

Table 4 shows that data extraction and modeling have significant complexities in three areas. First, the pre-processing phase requires the use of Selenium along with an undetected Chrome driver for data extraction and georeferencing via Google Maps, resulting in a high temporal cost. Second, modeling involves using algorithms such as Fuzzy C-Means and metaheuristic methods (such as PSO), whose complexity and execution time vary according to the number of dwellings, clusters and iterations. However, their scalability allows an improved performance as data increases. Finally, system failures can arise from a lack of data or a cold start, which limits the ability to identify behavioral patterns and decreases the effectiveness of predicting housing purchases.

It may fail as the result of a cold start or lack of data. There are some cases in the process in which other techniques have been used, subjectivity with natural language processing

is presented, and clustering techniques are applied in which clusters are highly heterogeneous.

In summary, the graph provides a comprehensive view of the performance of the model, highlighting its strengths, such as high accuracy and recall, and areas that could benefit from further improvements, such as diversity. Together, these metrics facilitate a comprehensive understanding of the model and provide a solid basis for future optimizations. Upon comparing these results with those of previous research, the model proves to have a competitive performance and promises significant improvements in user experience.

VI. CONCLUSION

One of the fundamental challenges in the field of adequate housing is that of ensuring a reliable and safe environment in which to recommend housing according to user preferences. This information is essential for stakeholders in order to provide housing that meets habitability standards and satisfies residents' needs. With the continuous increase in demand for adequate housing, this becomes a significant challenge. This paper proposes a cost-effective and practical approach with which to address these objectives in this context.

In this study, the performance of a fuzzy housing recommendation model has been evaluated using a radar graph to represent four key metrics: precision, recall, diversity and user satisfaction. The results revealed high precision (0.816), suggesting that most recommendations are relevant, along with moderately high recall (0.89), indicating that the most appropriate items are retrieved. Moderate diversity (0.175) was also observed, reflecting a reasonable variety of recommendations, while user satisfaction scored 0.58, which could be improved by means of additional testing with collaborative filtering.

The project focuses on R&D&I (Research, Development, and Innovation), advancing knowledge by analyzing recommendation systems and developing an innovative model for housing recommendations. This model combines distances to key spatial points with latent housing characteristics, thus providing a more comprehensive assessment of the environment. It additionally handles the uncertainty in user preferences through the use of fuzzy logic, an underexplored approach that enhances the personalization of recommendations. The system optimization by means of the bat algorithm reduces search times, and the use of a hybrid system that combines collaborative and content-based filtering increases the accuracy and relevance of recommendations.

The primary goal of this study was to develop a fuzzy housing recommendation model, and although significant progress has been made, there is still more to be done. It is recommended that more data be collected so as to include other elements regarding adequate housing, as per UN standards, and incorporate probabilistic expert systems and spatiotemporal models. These actions would improve the accuracy and relevance of recommendations, thus ensuring a satisfactory experience for users. The results obtained, such

as the solid performance of key metrics (precision, recall, diversity) and a validation with real users, align closely with the objectives of the study, and the aim of continually improving user satisfaction and system robustness that are consistent with the outcomes observed and the methodological framework. Furthermore, the practical implementation of the model in a web system under an MVC architecture allows validation and real-world use, providing reassurance about the applicability of the model and underscoring the innovative nature of the project and its significant contribution to the field of recommendation systems and urban planning.

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